2.24 Differentiation: The Product Rule

Worked Solutions 1–23

1 a)

$\frac{dy}{dx} = x \times 2(x-1) + (x-1)^2 = 2x(x-1) + (x-1)^2$

1 b)

$\frac{dy}{dx} = x^2 \times 4(2x+7) + 2x \times (2x+7)^2 = 4x^2(2x+7) + 2x(2x+7)^2$

1 c)

$\frac{dy}{dx} = 4x \times -4(7-x)^3 + 4(7-x)^4 = -16x(7-x)^3 + 4(7-x)^4$

1 d)

$\frac{dy}{dx} = 9(x+2)^2(3x+1)^2 + 2(x+2)(3x+1)^3$

1 e)

$$\frac{dy}{dx} = (3-x)^2 \times 2 \times 3(2x+9)^2 - 2(3-x)(2x+9)^3 = 6(3-x)^2(2x+9)^2 - 2(3-x)(2x+9)^3$$

1 f)

$$\begin{aligned}\frac{dy}{dx} &= \sqrt{2x+5} \times 8(4x+1) + 2 \times \tfrac{1}{2}(2x+5)^{-1/2}(4x+1)^2 \\ &= 8\sqrt{2x+5}(4x+1) + (2x+5)^{-1/2}(4x+1)^2\end{aligned}$$

1 g)

$$\frac{dy}{dx} = (\tfrac{1}{x} - 2x)(-3x^{-4} - 10x^{-6}) + (-x^{-2} - 2)(\tfrac{1}{x^3} + \tfrac{2}{x^5})$$

2)

Student A: incorrectly multiplied out the brackets. Incorrectly differentiated the x^5 term.

Student B: Correct method.

Student C: incorrectly multiplied the two differentials of the two brackets together without applying the product rule.

3 a)

$\frac{dy}{dx} = x \times \frac{1}{x} + \ln x = 1 + \ln x$

3 b)

$\frac{dy}{dx} = x^{1/2} \sec^2 x + \frac{1}{2}x^{-1/2} \tan x$

3 c)

$\frac{dy}{dx} = \sin x \times -\sin x + \cos x \cos x = \cos^2 x - \sin^2 x$

3 d)

$\frac{dy}{dx} = \frac{1}{x}\cos x - \sin x \ln x$

3 e)

$\frac{dy}{dx} = 2^x \cos x + \ln 2 \times 2^x \sin x$

3 f)

$\frac{dy}{dx} = \frac{x^2+1}{x} + 2x \ln x$

4 a)

$\frac{dy}{dx} = xe^x + e^x = e^x(x+1)$

4 b)

$\frac{dy}{dx} = (\ln 4)x^2 4^x + 2x \times 4^x = 4^x x(x \ln 4 + 2)$

4 c)

$\frac{dy}{dx} = e^x \sec^2 x + e^x \tan x = e^x(\sec^2 x + \tan x)$

4 d)

$\frac{dy}{dx} = 2^x \cos x + \ln 2 \times 2^x \sin x = 2^x(\cos x + \ln 2 \sin x)$

4 e)

$$\frac{dy}{dx} = (x^2 - 12) \times -2e^{-2x} + 2xe^{-2x} = -2e^{-2x}(x^2 - 12) = -2e^{-2x}(x-4)(x+3)$$

4 f)

$\frac{dy}{dx} = 2\cos 2x\ e^{4x} + \sin 2x \times 4e^{4x} = 2e^{4x}(\cos 2x + 2\sin 2x)$

5 a)

$\frac{dy}{dx} = 2x^3(x+2) + 3x^2(x+2)^2$. When $x = 1$, $y = 9$.

$$\left.\frac{dy}{dx}\right|_{x=1} = 6 + 27 = 33$$

$y - 9 = 33(x-1) \Rightarrow y = 33x - 24$.

5 b)

$\frac{dy}{dx} = \frac{3}{2}x^5(3x+10)^{-1/2} + 5x^4\sqrt{3x+10}$. When $x = 2$, $y = 128$.

$$\left.\frac{dy}{dx}\right|_{x=2} = 332$$

$y - 128 = 332(x-2) \Rightarrow y = 332x - 536$.

5 c)

$\frac{dy}{dx} = (x^2 - x + 4)e^x + (2x - 1)e^x$. When $x = 0$, $y = 4$ so $\left.\frac{dy}{dx}\right|_{x=0} = 4 - 1 = 3$. $y - 4 = 3(x - 0) \Rightarrow y = 3x + 4$.

6 a)

$2x^2[2 + (x + 1)] = 2x^2(x + 3)$

6 b)

$(2x - 1)^4[3(2x - 1) - (x + 1)] = (2x - 1)^4[6x - 3 - x - 1] = (2x - 1)^4(5x - 4)$

6 c)

$(3x + 1)^2[4(3x + 1) + (x + 2)] = (3x + 1)^2[12x + 4 + x + 2] = (3x + 1)^2(13x + 6)$

6 d)

$\frac{1}{3}(6x + 5)^2[(6x + 5) + 3(x - 1)] = \frac{1}{3}(6x + 5)^2[6x + 5 + 3x - 3] = \frac{1}{3}(6x + 5)^2(9x + 2)$

6 e)

$$\frac{1}{2(1-x)^5} + \frac{4}{(1-x)^6} = \frac{1}{2(1-x)^6}[(1-x)+8] = \frac{1}{2(1-x)^6}(9-x) = \frac{9-x}{2(1-x)^6}$$

6 f)

$(3x-1)(4x+1)^2[4(4x+1)+9(3x-1)] = (3x-1)(4x+1)^2[16x+4+27x-9] = (3x-1)(4x+1)^2(43x-5)$

7 a)

$$(x+1)^2[2(x+1)+(x-1)] = 0 \Rightarrow (x+1)^2[2x+2+x-1] = 0 \Rightarrow (x+1)^2(3x+1) = 0$$

Therefore $x = -1$, $x = -\frac{1}{3}$.

7 b)

$$\frac{1}{3(2x+1)^2} + \frac{1}{(2x+1)^3} = 0 \Rightarrow \frac{1}{3(2x+1)^3}[(2x+1)+3] = 0 \Rightarrow \frac{2x+4}{3(2x+1)^3} = 0$$

Therefore $x = -2$.

7 c)

$$(7x-1)^2(x-1)[5(7x-1) - 3(x-1)] = 0$$
$$(7x-1)^2(x-1)[35x - 5 - 3x + 3] = 0 \Rightarrow (7x-1)^2(x-1)(32x-2) = 0$$

Therefore $x = \frac{1}{7}$, 1, $\frac{1}{16}$.

8 a)

$\frac{dy}{dx} = 6x^3(3x - 5) + 3x^2(3x - 5)^2$. Stat pt when $\frac{dy}{dx} = 0$.

$$6x^3(3x-5) + 3x^2(3x-5)^2 = 0 \Rightarrow 3x^2(3x-5)[2x + (3x-5)] = 0$$
$$3x^2(3x-5)[5x-5] = 0$$

Therefore $x = 0$, $\frac{5}{3}$ or1 so $x = 1$. When $x = 1$, $y = 4$ so A is $(1, 4)$.

8 b)

$(0,0)$ and $(\frac{5}{3},0)$

8 c)

$0 < y < 4$

9 a)

$\frac{dy}{dx} = 7x(x+3)^6 + (x+3)^7 = (x+3)^6[7x + (x+3)] = (x+3)^6(8x+3)$

9 b)

$$\begin{aligned}\frac{dy}{dx} &= 8(x-3)(2x+1)^3 + (2x+1)^4 \\ &= (2x+1)^3[8(x-3) + (2x+1)] \\ &= (2x+1)^3[8x - 24 + 2x + 1] = (2x+1)^3(10x - 23)\end{aligned}$$

9 c)

$$\begin{aligned}\frac{dy}{dx} &= 40x(2x+1)^3 + 5(2x+1)^4 = (2x+1)^3[40x + 5(2x+1)] \\ &= (2x+1)^3(50x+5) = 5(2x+1)^3(10x+1)\end{aligned}$$

9 d)

$$\begin{aligned}\frac{dy}{dx} &= 2(1+x^2)e^{2x} + 2xe^{2x} = e^{2x}[2(1+x^2) + 2x] \\ &= e^{2x}[2x^2 + 2x + 2] = 2e^{2x}(x^2 + x + 1)\end{aligned}$$

10 a)

$\frac{dy}{dx} = -8000xe^{-2x} + 4000e^{-2x} = -4000e^{-2x}(2x-1)$. Stat pt when $\frac{dy}{dx} = 0$ so $x = \frac{1}{2}$. So $(\frac{1}{2}, \frac{2000}{e})$.

10 b)

$$\begin{aligned}\frac{d^2y}{dx^2} &= -8000e^{-2x} + 8000e^{-2x}(2x-1) \\ \left.\frac{d^2y}{dx^2}\right|_{x=1/2} &= -8000e^{-1} < 0 \text{ so maximum}\end{aligned}$$

11 a)

150employees

11 b)

$\frac{dy}{dt} = -\frac{\pi}{2}t\cos\frac{\pi}{6}t - 3\sin\frac{\pi}{6}t$

11 c)

$\left.\frac{dy}{dt}\right|_{t=3} = -\frac{3\pi}{2}\cos\frac{\pi}{2} - 3\sin\frac{\pi}{2} = -3$

11 d)

On exactly1st April the company is losing3 employees per day.

12)

$\frac{dy}{dx} = 2x^4e^{2x} + 4x^3e^{2x}$.

$$\left.\frac{dy}{dy}\right|_{x=1} = 2e^2 + 4e^2 = 6e^2$$

When $x = 1$, $y = e^2$.

$$y - e^2 = 6e^2(x-1) \Rightarrow y - e^2 = 6e^2x - 6e^2 \Rightarrow y - 6e^2x + 5e^2 = 0$$

13)

$\frac{dy}{dx} = x \times \frac{1}{x} + \ln x = 1 + \ln x$. Stat pt when $\frac{dy}{dx} = 0$ so $1 + \ln x = 0 \Rightarrow \ln x = -1 \Rightarrow x = e^{-1}$. So $(e^{-1}, -e^{-1})$.

$$\frac{d^2y}{dx^2} = \frac{1}{x} \text{ so } \left.\frac{d^2y}{dx^2}\right|_{x=e^{-1}} = e > 0 \text{ so a minimum}$$

14 a)

$\frac{dy}{dx} = e^{-x}\cos x - e^{-x}\sin x = e^{-x}(\cos x - \sin x)$. At $x = \frac{\pi}{4}$,

$$\frac{dy}{dx} = e^{-\pi/4}(\cos\tfrac{\pi}{4} - \sin\tfrac{\pi}{4}) = e^{-\pi/4}(\tfrac{\sqrt{2}}{2} - \tfrac{\sqrt{2}}{2}) = 0$$

So stat pt at $x = \frac{\pi}{4}$.

$$e^{-\pi/4}\sin\tfrac{\pi}{4} = \tfrac{\sqrt{2}}{2}e^{-\pi/4} \text{ so } (\tfrac{\pi}{4}, \tfrac{\sqrt{2}}{2}e^{-\pi/4}) \text{ as req.}$$

14 b)

$$\begin{aligned}\frac{d^2y}{dx^2} &= e^{-x}(-\sin x - \cos x) - e^{-x}(\cos x - \sin x)\\ &= e^{-x}(-\sin x - \cos x - \cos x + \sin x) = -2e^{-x}\cos x\end{aligned}$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=\pi/4} = -2e^{-\pi/4}\cos\tfrac{\pi}{4} = -\sqrt{2}e^{-\pi/4} < 0 \text{ so max}$$

15 a)

$$\frac{dy}{dx} = -10e^x\sin x + 10e^x\cos x = 10e^x(\cos x - \sin x)$$

Stat pt when $\frac{dy}{dx} = 0$.

$$10e^x(\cos x - \sin x) = 0 \text{ but } e^x \neq 0 \Rightarrow \cos x - \sin x = 0 \Rightarrow \sin x = \cos x \Rightarrow \tan x = 1$$

15 b)

For $x \geq 0$, stat pts at $x = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \ldots$.

$$y = \ldots, 5\sqrt{2}e^{\pi/4}, 5\sqrt{2}e^{5\pi/4}, 5\sqrt{2}e^{9\pi/4}, \ldots$$

Therefore $r = -e^{\pi}$.

16 a)

$\mathrm{f}'(x) = 6x^2 e^{10x^2}\cos(2x^3) + 20xe^{10x^2}\sin(2x^3)$

16 b)

$$\begin{aligned}\mathrm{f}'(x) &= 2x \times 5^{x^2}\sin(1-x^2) + 2x\ln 5 \times 5^{x^2}\cos(1-x^2)\\ &= 2x5^{x^2}\sin(1-x^2) + 2x(\ln 5)5^{x^2}\cos(1-x^2)\end{aligned}$$

16 c)

$\mathrm{f}'(x) = \frac{2x}{x^2}(x^2+1)^3 + 6x(x^2+1)^2\ln(x^2) = \frac{2}{x}(x^2+1)^3 + 6x(x^2+1)^2\ln(x^2)$

17)

$$\frac{dy}{dx} = \ln 2 \times 6x \times 2^{3x^2}$$

$$\frac{d^2y}{dx^2} = \ln 2 \times 36x^2 \times 2^{3x^2} + 6\ln 2 \times 2^{3x^2}$$

So $\frac{d^2y}{dx^2} > 0$ for all x since $x^2 \geq 0$. Therefore convex for all x.

18)

$\frac{dy}{dx} = \frac{5-2x}{x+1} - 2\ln(x+1)$. Stat pt when $\frac{dy}{dx} = 0$.

$$\left.\frac{dy}{dx}\right|_{x\approx 1.04} = \frac{5-2\times 1.04}{1.04+1} - 2\ln(1.04+1) = 0.0005540\ldots$$

$$\left.\frac{dy}{dx}\right|_{x\approx 0.94} = \frac{5-2\times 0.94}{0.94+1} - 2\ln(0.94+1) > 0$$

$$\left.\frac{dy}{dx}\right|_{x\approx 1.14} = \frac{5-2\times 1.14}{1.14+1} - 2\ln(1.14+1) < 0$$

So the gradient before the point is positive and the gradient after the point is negative.

19 a)

$\frac{dy}{dx} = x^2e^x + 2xe^x = xe^x(x+2)$. Stat pt when $\frac{dy}{dx} = 0$ so $xe^x(x+2) = 0$. $e^x \neq 0$ therefore $x = 0$, $x = -2$. $(0,0)$ and $(-2, 4e^{-2})$.

19 b)

$$\begin{aligned}\frac{dy}{dx} &= 30x(3x-1)^4(3x^2+1)^4 + 12(3x-1)^3(3x^2+1)^5 \\ &= 6(3x-1)^3(3x^2+1)^4[5x(3x-1)+2(3x^2+1)] \\ &= 6(3x-1)^3(3x^2+1)^4[15x^2-5x+6x^2+2] \\ &= 6(3x-1)^3(3x^2+1)^4(21x^2-5x+2)\end{aligned}$$

Stat pt when $\frac{dy}{dx}=0$ so $6(3x-1)^3(3x^2+1)^4(21x^2-5x+2)=0$. For $3x^2+1=0$, there are no real solutions (as $x^2\geq 0$, so $3x^2+1\geq 1$). For $21x^2-5x+2=0$, $b^2-4ac=-143<0$ so no real solutions. So the only solution comes from $3x-1=0\Rightarrow x=\frac{1}{3}$. Therefore there is a stationary point at $(\frac{1}{3},0)$.

19 c)

$$\begin{aligned}\frac{dy}{dx} &= 2\sqrt{x}(x-1)+\tfrac{1}{2}x^{-1/2}(x-1)^2 \\ &= 2\sqrt{x}(x-1)+\frac{1}{2\sqrt{x}}(x-1)^2 \\ &= \frac{4x}{2\sqrt{x}}(x-1)+\frac{1}{2\sqrt{x}}(x-1)^2 \\ &= \frac{1}{2\sqrt{x}}(x-1)[4x+(x-1)] = \frac{1}{2\sqrt{x}}(x-1)(5x-1)\end{aligned}$$

Stat pt when $\frac{dy}{dx}=0$ so $\frac{1}{2\sqrt{x}}(x-1)(5x-1)=0$. So $x=1$, $x=\frac{1}{5}$ therefore $(1,0)$ and $(\frac{1}{5},\frac{16}{25\sqrt{5}})$.

20 a)

$\frac{dx}{dy}=\frac{3(y+2)}{3y-5}+\ln(3y-5)$

20 b)

When $x=0$, $0=(y+2)\ln(3y-5)$ so $y=-2$ or $3y-5=1\Rightarrow y=2$.

$$\left.\frac{dx}{dy}\right|_{y=-2} = \ln(-11) \text{ not possible}$$

$$\left.\frac{dx}{dy}\right|_{y=2} = 12 \text{ so } \frac{dy}{dx}=\tfrac{1}{12}$$

21)

$y=x^2(\tan x)^3$.

$$\frac{dy}{dx} = 3x^2\sec^2 x(\tan x)^2 + 2x(\tan x)^3$$

$$\left.\frac{dy}{dx}\right|_{x=\pi/4} = 3\frac{\pi^2}{16}\times 2 + 2\times\frac{\pi}{4} = \frac{3\pi^2}{8}+\frac{\pi}{2} = \frac{\pi}{8}(3\pi+4) \text{ as req.}$$

22 a)

When $t = 0$, $h = 10 + 25 \times 2 = 60$ m.

22 b)

$$\frac{dh}{dt} = -\frac{\pi}{3} \times 25e^{-0.05t}\sin(\tfrac{\pi}{3}t) - 1.25e^{-0.05t}(1 + \cos(\tfrac{\pi}{3}t))$$

$$\left.\frac{dh}{dt}\right|_{t=3} = 0$$

22 c)

When $t = 3$, $h = 10$ so the bungee jumper has reached the minimum height which is10m above the ground.

23 a)

$\mathrm{f}'(x) = 2x(4x+3)^{-1/2} + \sqrt{4x+3}$

23 b)

$$\begin{aligned}
\mathrm{f}''(x) &= -4x(4x+3)^{-3/2} + 2(4x+3)^{-1/2} + 2(4x+3)^{-1/2} \\
&= -4x(4x+3)^{-3/2} + 4(4x+3)^{-1/2} \\
&= -\frac{4x}{(4x+3)^{3/2}} + \frac{4}{(4x+3)^{1/2}} \\
&= -\frac{4x}{(4x+3)^{3/2}} + \frac{4(4x+3)}{(4x+3)^{3/2}} \\
&= \frac{-4x+16x+12}{(4x+3)^{3/2}} = \frac{12(x+1)}{(4x+3)^{3/2}} \text{ as req.}
\end{aligned}$$

23 c)

Stat pt when $\mathrm{f}'(x) = 0$ so at $x = -\frac{1}{2}$. $\mathrm{f}''(-\frac{1}{2}) = 6 > 0$ so minimum point.